MIL-C-81214A(AS)
1 January 1968
Superseding
MIL-C-81214(WP)
15 February 1965

MILITARY SPECIFICATION

CHAFF, COUNTERMEASURES RR-124( )/AL

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

*1.1 Scope - The Radar Reflector Package covered by this specification shall consist of a plastic container loaded with aluminum coated glass chaff. The package will be dispensed from an AN/ALE-18 Chaff Dispenser Set to produce confusing signals in radar systems operating in the frequency range of 2,500-10,500 MHz.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triple Wall
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock

Military

MIL-B-131	Barrier Material: Water Vaporproof, Flexible
MIL-B-117	Bags, Sleeves, and Tubing - Interior Packaging
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aeronautical and Associated Equipment

FSC5895

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MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

2.1.1 Availability of Documents

(1) When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration.

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*3.2.1 Selection of Materials - Specifications and Standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143.

*3.2.2 Material - The dipole material shall be Type E glass fiber, coated with 2-4 micron thickness aluminum alloy of 99.4% purity or greater. A minimum of 90% of the surface of the glass shall be so coated. A secondary slip coating shall be applied over the metal coating to minimize cohesion between adjacent dipoles. The metal coating shall provide an electrically conductive path which is continuous from one end of the fiber to the other. The direct current resistance of the metal coat shall not exceed ten ohms per inch. One dipole selected at random from each of the five units of chaff in every sampled chaff round (see paragraph 4.4.2) shall be examined for compliance with these requirements by means of repeatable optical and electrical measurements.

*3.3 Design and Construction - The chaff rounds shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient ($25^{\circ}C \pm 5^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

*3.3.2 Service Conditions - The chaff rounds shall operate satisfactorily under any of the environmental service conditions or reasonable combinations of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

*3.3.3 Identification of Product - Nomenclature assignment shall be in accordance with MIL-N-18307. Identification shall conform to MIL-STD-130 and be effected by marking the cover of the chaff assembly with the legend: "Chaff, RR124/ ALE 18, S, C, X-Band FSN5895-911-7831-NA08."

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*3.4 Performance - The chaff rounds to be supplied hereunder shall be suitable for dispensing from an AN/ALE-18 Chaff Dispensing Set. The chaff shall produce a chaff echo in radar systems within the frequency range of 2,500-10,500 MHz. The individual chaff round shall simulate an airborne vehicle whose mean value of radar cross-section is a minimum of: 100 square meters within the frequency range of 2,500-3,500 MHz; 75 square meters within the frequency range of 4,500-6,500 MHz; and 50 square meters within the frequency range of 8,300-10,500 MHz. The chaff is required to attain these values of cross-section two seconds after ejection in at least eight out of ten trials, at the resonant frequency within each of the aforementioned frequency bands.

*3.5 Detail Requirements -

*3.5.1 Chaff - The chaff rounds supplied hereunder shall meet the following requirements.

*3.5.1.1 Form Factor - The chaff dipoles shall be assembled in individual units and inserted into the chaff container in five layers, as shown in Figure 1. The dipole chaff unit length and the number of chaff units required per chaff load for each of the five layers are shown in Table I. The dimensions of the dipole cross-section (at right angles to the longitudinal axis) and the number of dipoles required per chaff unit are shown in Table II. The longitudinal cumulative tolerance of each chaff layer shall not exceed a positive .08 inch for layers 1 and 2 and .04 inch for layers 3, 4 and 5.

TABLE I

Chaff Units per Load	Chaff Unit Length (inches)
3rd and 5th Layers	
6	.61 ± .02
1st, 2nd and 4th Layers	
1	1.99 ± .02
1	1.13 ± .02
1	.58 ± .02

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TABLE II

Cross-Section Geometry	Cross-Section Dimensions (inches)	Numbers of Dipoles per Chaff Unit
Circular	0.001 ± 20% (dia.)	Layers 1, 2, 3, and 4 270,000 ± 5% Layer 5 110,000 ± 5%

*3.5.2 Chaff Round -

*3.5.2.1 Function - The chaff round shall be completely compatible with and suitable for dispensing of the chaff payload from an AN/ALE-18 dispenser. The chaff round shall provide a gas seal and retainer (cover) for the chaff load capable of normal handling including repeated insertion and removal from the AN/ALE-18 dispenser. The chaff round shall be designed to release all components except the container itself when used with the above dispenser. The chaff round shall also be designed such that its use with the AN/ALE-18 dispenser shall preclude any damage, or scoring of the AN/ALE-18 dispenser or premature release of chaff under the service conditions specified in paragraph 3.3.2.

*3.5.2.2 Form Factor - The form factor for the chaff container shall be compatible with its intended use as specified in paragraph 3.5.2.1 above and shall be subject to the approval of the procuring activity.

*3.5.2.3 Contents - The chaff round shall consist of a container, cover and gas seal and such other components as are necessary to accomplish the functions described in paragraph 3.5.2.1 above.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

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(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests

*4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on ninety chaff rounds representative of the production units to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of para. 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

*4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the chaff loads meet all the requirements of this specification and the contract. Preproduction tests shall include one complete cycle of the humidity test described in paragraph 4.4.2 of MIL-T-5422 and the rough handling test referenced in paragraph 5.2.1 of this specification.

*4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production chaff loads shall be delivered prior to the approval of the preproduction sample. Prefabrication of production chaff loads prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of chaff loads to be submitted for acceptance. The preproduction sample shall not be considered as chaff loads under the contract.

*4.2.4 Production Chaff Rounds - Chaff rounds supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each chaff load shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and, for units already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring agency.

*4.3 Initial Production Tests - One hundred twenty of the first one thousand chaff rounds shall be selected and sent at the contractor's expense to a designated

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Government laboratory for tests. These chaff rounds shall be selected by the procuring activity after the chaff rounds have successfully passed all individual tests. The preproduction sample shall not be selected for this test.

*4.3.1 Scope of Tests - These chaff rounds may be subjected to any and all tests the procuring activity deems necessary to assure that the production chaff loads are equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality, and that they meet all applicable requirements.

*4.3.2 Accessory Material - In addition to the chaff rounds submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the chaff loads.

*4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on chaff loads previously accepted when requested by the procuring activity.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Special Tests

*4.4.1 Individual Tests - Each chaff round submitted for acceptance may be subjected to an individual test. This test, an examination of product, shall be adequate to determine compliance with the requirements of material, workmanship, and operational adequacy.

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*4.4.2 Sampling Tests - Chaff rounds selected for sampling tests shall first have passed the individual tests. One out of every one thousand chaff rounds shall be selected for sampling tests by the Government Inspector.

*4.4.2.1 Scope of Tests

(1) Sampling tests shall include all tests deemed necessary by the procuring activity to determine that the chaff rounds meet all the requirements of this specification and the contract. Sampling tests shall include optical and electrical measurements to insure compliance of the product with the requirements specified in paragraph 3.2.2, and environmental tests in accordance with paragraph 4.2.2.

(2) Sampling tests of chaff rounds shall also include operational tests to be conducted following environmental tests of the sample chaff rounds. The Operational Tests shall include live firings from an AN/ALE-18 Chaff Dispenser to demonstrate compliance with the functional requirements of paragraph 3.5.2.1 and overall effectiveness of the payload including satisfactory discharge and separation of the dipoles, retention and non-destruction of the chaff container (except the cover and gas seal) and ease of removal of the chaff container after firing.

*4.4.3 Special Tests - Special tests shall be conducted on a quantity of chaff rounds for the prupose of checking the effect of any design or material change on the performance of the chaff rounds and to assure adequate quality control. The chaff rounds selected for special tests may be selected from chaff rounds previously subjected to the sampling tests.

*4.4.3.1 Special Test Schedule - Selection of chaff rounds for special tests shall be made as follows:

(1) On early chaff rounds after an engineering or major change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

*4.4.3.2 Scope of Tests - Special tests shall consist of such tests as required by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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*4.4.4 Chaff Load Failure - Should a failure occur during either the sampling or special tests, the following actions shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all chaff loads for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Government-Loaned Property - The procuring activity shall provide the contractor, on a loan basis, the quantity of AN/ALE-18 Chaff Dispensers required to perform the quality assurance provisions of this specification.

*4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

5. PREPARATION FOR DELIVERY

*5.1 Packaging of Chaff Rounds in Inner Carton - Forty-eight chaff rounds shall be packaged per carton. Each round shall be separated by a laminated board filler or egg crating conforming to PPP-F-320. The filler material shall not be crushed or deformed in handling, storing, or in the conduct of the test specified in paragraph 5.2.1. The filler shall be designed to support the chaff round such that it is not damaged during normal handling or in the conduct of the tests specified in paragraph 5.2.1.

*5.1.1 Marking of Chaff Units - See paragraph 3.3.3.

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*5.1.2 Carton Details - The unit container shall be a folding box (telescope type, half slotted) conforming to PPP-B-636, Type I, Class 2, Grade 3 (V3c).

*5.1.3 Inner Carton - The inner carton containing the individual rounds of chaff shall be enclosed in a close-fitting heat-sealed bag, fabricated from barrier material conforming to MIL-B-131, Class 1. Bag to be manufactured in accordance with MIL-B-117, Type 1, Class E. The air volume within the barrier material shall be reduced to a practicable minimum before sealing the barrier.

*5.2 Packing - The inner cartons shall be packed in an exterior fiberboard package for overseas shipment in accordance with PPP-B-640, Class 2, Grade A. The exterior dimensions shall not exceed two feet in height, width or depth.

*5.2.1 Preproduction Tests - Preproduction tests of a loaded exterior fiberboard package shall be performed in accordance with the requirements of paragraphs 10.4 and 10.5 in Appendix C of MIL-STD-794, except that preproduction chaff rounds shall be used in lieu of dummy rounds.

*5.3 Marking of Shipments - Interior cartons and exterior packaging shall be marked in accordance with MIL-STD-129 and include the legend "CHAFF, Countermeasures RR-124/ALE-18 for S,C,X-Band FSN 5895-911-7831-NA-08".

6. NOTES

*6.1 Ordering Data - Procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See para. 5.1).

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

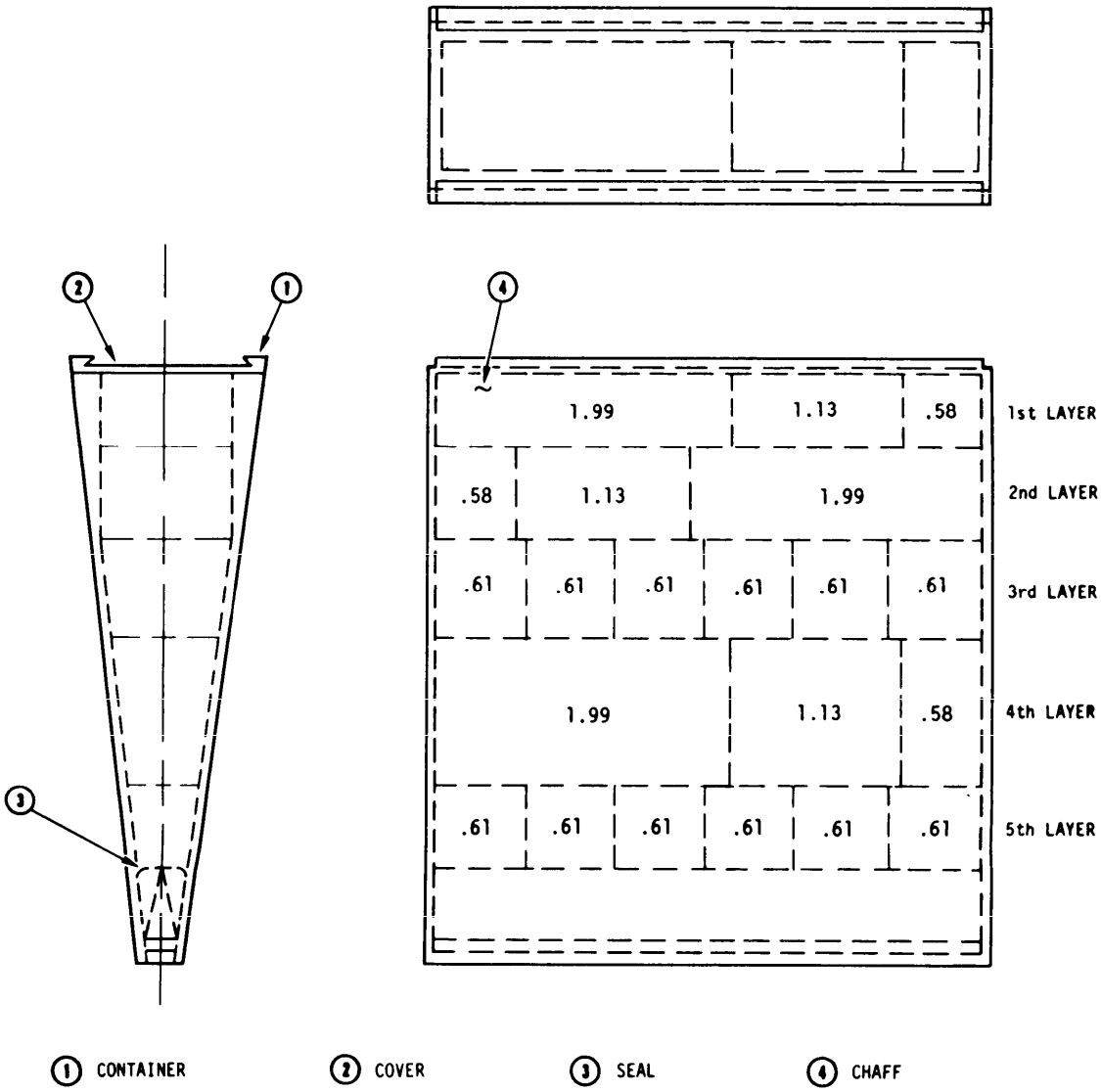
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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.3 Type Designations - The parentheses (*) when used in the equipment type designations, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with para. 3.3.3. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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CHAFF ROUNDS (ASSEMBLY)
FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81214A(AS) CHAFF, COUNTERMEASURES RR-124( )/AL

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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